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Device and method for blocking current using disconnecter

Abstract

An energy storage system (ESS) including a plurality of battery modules and a battery protection unit, the battery protection unit includes a microcontroller (MCU) for controlling charge and discharge of the ESS, a first main contactor and a second main contactor connecting or disconnecting between the plurality of battery modules and an output terminal of the ESS under control of the MCU, and a disconnecter disposed between the first main contactor and the second main contactor and connecting or disconnecting the plurality of battery modules, the first main contactor, and the second main contactor, the first main contactor and the second main contactor are turned on or off, by the MCU, depending on whether a predetermined voltage is applied.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a device and method for cutting-off current using a disconnecter in an ESS.

BACKGROUND ART

(2) An Energy Storage System (ESS) is used to stably connect a power produced from renewable energy to a power system. Renewable energy is electric energy generated through solar or wind power generation, and has a disadvantage in that power production stability is poor. In order to solve these shortcomings, renewable energy is used in combination with the ESS. The ESS is a battery using Li-Ion Battery (LIB), and the like, and stores excess power and discharges it when power is insufficient to stabilize power supply and demand. In other words, the ESS improves power utilization efficiency by accumulating and storing renewable energy that is insignificantly gathered when power demand is low, and then using the stored power during peak times when the power demand is high.

(3) Such an ESS requires a function of manually cutting off the current by the user for abnormal situations or inspection.

(4) Conventionally, a circuit breaker was used for this function.

(5) However, since the conventional circuit breaker further includes a function of blocking the current path in case of overcurrent in addition to the function of manually blocking the current by the user, there is a disadvantage of high price.

(6) On the other hand, it is possible to manually cut off the current by using a disconnecter instead of an expensive circuit breaker but the disconnecter can be turned on or off only in the no-load state, so that there is a disadvantage that it cannot be turned on or off when the ESS is in use.

(7) Accordingly, the present invention proposes a device and method for manually cutting off current even if it is not in a no-load state by using an inexpensive disconnecter instead of such a conventional expensive circuit breaker. (Patent Document 0001) Korean Laid-Open Patent Publication No. 10-2015-0090371

DISCLOSURE

Technical Problem

(8) The present invention provides a device and method capable of manually blocking a current flowing through an ESS by using a disconnecter even in a no-load state.

Technical Solution

(9) According to an embodiment of the present invention, an energy storage system (ESS) includes

a plurality of battery modules and a battery protection unit, wherein the battery protection unit includes: a microcontroller (MCU) configured to control charge and discharge of the ESS; a first main contactor and a second main contactor connecting or disconnecting between the plurality of battery modules and an output terminal of the ESS under control of the MCU; and a disconnecter disposed between the first main contactor and the second main contactor and connecting or disconnecting the plurality of battery modules, the first main contactor, and the second main contactor.

(10) The first main contactor and the second main contactor are turned on or off, by the MCU, depending on whether a predetermined voltage is applied.

(11) The disconnecter may include: a first current cut-off switch having a first end connected to a positive (+) end of the plurality of battery modules and a second end connected to a first main contactor to block a current path flowing through the first main contactor; and a second current cut-off switch having a first end connected to a negative (−) end of the plurality of battery modules and a second end connected to a second main contactor to block a current path flowing through the second main contactor.

(12) The disconnecter may include an auxiliary switch for controlling whether a predetermined voltage is applied to the first main contactor and the second main contactor.

(13) When the auxiliary switch is turned off, the first main contactor and the second main contactor may be turned off by cutting off predetermined power supplied to the first main contactor and the first main contactor, wherein when the first main contactor and the second main contactor are turned off, the ESS may become a no-load state.

(14) The first and second current cut-off switches may be turned off after the first and second main contactors are turned off.

(15) The battery protection unit may further include: a first fuse having a first end connected to a positive (+) end of the plurality of battery modules and a second end connected to the first main contactor through the disconnecter; and a second fuse having a first end connected to a negative (−) end of the plurality of battery modules and a second end connected to the second main contactor through the disconnecter.

(16) When at least one of overvoltage, overcurrent, and high temperature is detected, the MCU may turn off the first main contactor and the second main contactor.

(17) The ESS may further include a pre-charge FET connected in parallel to the first main contactor to prevent inrush current.

(18) A method for performing manual power disconnection in an energy storage system (ESS) including a disconnecter according to an embodiment of the present invention includes: a main contactor off step of converting a rear end of the disconnecter into a no-load state; and a current path blocking step of blocking an output current path of the ESS using the disconnecter after performing the main contactor off step.

(19) The main contactor off step may turn off the main contactor by cutting off the power supplied to the main contactor, thereby converting the rear end of the disconnecter into a no-load state.

(20) The disconnecter includes a first current cut-off switch and a second current cut-off switch. The current path blocking step may turn off the first current cut-off switch and the second current cut-off switch of the disconnecter to block the output current path of the ESS after the main contactor off step is performed.

Advantageous Effects

(21) The present invention can manually cut off the current in the ESS without using an expensive circuit breaker.

(22) In addition, the present invention is provided with a circuit for converting the ESS into a no-load state so that a low-cost passive current cut-off device can be applied by applying a disconnecter in place of the conventional circuit breaker so that it is possible to provide a current cut-off device at low cost compared to the prior art.

Description

DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a diagram showing an ESS according to an embodiment of the present invention.

(2) FIG. 2 is a flowchart illustrating a method of manually blocking current in an ESS according to an embodiment of the present invention.

MODE FOR INVENTION

(3) Hereinafter, embodiments of the present invention will be described in detail with reference to the accompanying drawings so that those of ordinary skill in the art may easily implement the present invention. However, the present invention may be implemented in various forms and is not limited to the embodiments described herein. In the drawings, parts irrelevant to the description are omitted in order to clearly describe the present invention, and like reference numerals refer to like elements throughout the specification.

(4) 1. ESS According to Embodiment of Present Invention

(5) In an ESS, by detecting overcurrent in addition to the function to manually cut off the current.

(6) On the other hand, a disconnecter is used as another element for manually blocking the current.

(7) However, disconnectors have the advantage of being cheap, but the disconnecter's current cut-off switch can be turned on or off only when the disconnecter is in a no-load state, so that in general, there was a limit to apply and use in an ESS connected to a load.

(8) Accordingly, in the present invention, even if the disconnecter is not in a no-load state, a device for manually blocking current using a disconnecter has been developed.

(9) That is, in the present invention, as configuring an auxiliary switch in a circuit to which a disconnecter is connected, when manual current interruption is required, by controlling the auxiliary switch, the disconnecter is switched to a no-load state to manually cut off the current.

(10) FIG. 1 is a diagram showing an ESS according to an embodiment of the present invention.

(11) Hereinafter, an ESS according to an embodiment of the present invention will be described with reference to FIG. 1.

(12) An ESS **10** according to the embodiment of the present invention may be configured to include a battery protection unit **200**.

(13) 1) Battery Protection Unit **200** of Present Invention

(14) The battery protection unit **200** includes an MCU **230** that controls the charging and discharging of the ESS **10**, a first main contactor **221** and a second main contactor **222** connecting or blocking the plurality of battery modules **100** and the output terminals of the ESS under the control of the MCU **230**, and a disconnecter **240** that are provided between the first main contactor **221** and the second main contactor **222**, and connect or block the plurality of battery modules **10** and the first main contactor **221** and the second main contactor **222**.

(15) 1-1) MCU **230**

(16) When the ESS **10** needs to be charged or discharged, the MCU **230** according to an embodiment of the present invention charges or discharges the ESS by turning on the first main contactor **221** and the second main contactor **222**, and while the ESS **10** is charging or discharging, when at least one of overvoltage, overcurrent, and high temperature is detected, the MCU **230** may block charging or discharging of the ESS **10** by turning off the first main contactor **221** and the second main contactor **222**.

(17) 1-2) First Main Contactor **221** and Second Main Contactor **222**

(18) The first main contactor **221** and the second main contactor **222** may connect or short-circuit a plurality of battery modules and an output terminal of the ESS under the control of the above-described MCU **230**.

(19) For example, the first main contactor **221** is configured to connect between the (+) output terminal of the plurality of battery modules and the (+) output terminal of the ESS, and the second

main contactor **222** is configured to connect between the (-) output terminal of the plurality of battery modules and the (-) output terminal of the ESS.

(20) Meanwhile, the first main contactor **221** and the second main contactor **222** may be configured as relays that are turned on or off when a predetermined power is applied.

(21) During normal operation, the first main contactor **221** and the second main contactor **222** are controlled by the MCU **230** to connect or block the ESS charging or discharging path.

(22) In the present invention, in addition to this, depending on whether external power connected through the auxiliary switch **243** of the disconnecter **240** to be described later is applied, it has an additional function of turning on or off the first main contactor **221** and the second main contactor **222**, so that the disconnecter switches the ESS to a no-load state when the current is cut off

(23) 1-3) Disconnector **240**

(24) The disconnecter **240** is provided between the plurality of battery modules **100** and the first main contactor **221** and between the plurality of battery modules and the second main contactor **222** to cut off the output from the battery module.

(25) Such a disconnecter **240** includes a first current cut-off switch **241** for blocking the connection between the battery module and the first main contactor **221**, a second current cut-off switch **242** for blocking the connection between the battery module and the second main contactor **222**, and an auxiliary switch **243** for turning off the first main contactor **221** and the second main contactor **222** to allow the rear end of the disconnecter **240** to be in a no-load state.

(26) 1-3-1) First Current Cut-Off Switch **241** and Second Current Cut-Off Switch **242**

(27) Specifically, the first current cut-off switch **241** is connected between the (+) terminal of the plurality of battery modules **100** and the first main contactor **221**, thereby blocking the current path flowing through the first main contactor **221**.

(28) The second current cut-off switch **242** is connected between the (-) terminal of the plurality of battery modules **100** and the second main contactor **222**, thereby blocking the current path flowing through the second main contactor **222**.

(29) The output current of the disconnecter ESS is cut off through such cut-off control of the first and second switches.

(30) Meanwhile, the first current cut-off switch **241** and the second current cut-off switch **242** may be configured as elements that are turned on or off only when the disconnecter **240** is in a no-load state.

(31) In the present invention, in order to turn on or off the first current cut-off switch **241** and the second current cut-off switch **242**, the first main contactor **221** and the second main contactor **222** are turned off to switch the disconnecter to the no-load state.

(32) 1-3-2) Auxiliary Switch **243**

(33) In order to convert the disconnecter **240** into a no-load state, the first main contactor **221** and the second main contactor **222** are turned off by using the auxiliary switch **243**.

(34) Specifically, the auxiliary switch **243** may be configured to include a rotating handle. As the handle rotates, the auxiliary switch may be turned on or off.

(35) For example, if the ESS **10** is in a normal state, the auxiliary switch **243** is operated in an on state and supplies a predetermined voltage to the first main contactor **221** and the second main contactor **222**, and the first main contactor **221** and the second main contactor **222** are turned on, so that a first current path in which the first fuse **211**.fwdarw.first current cut-off switch **241**.fwdarw.first main contactor **221** are connected, is formed and a second current path in which the second fuse **212**.fwdarw.second current cut-off switch **242**.fwdarw.second main contactor **222** are connected, is formed.

(36) On the other hand, when an abnormal operation of the ESS occurs or an inspection of the ESS is required, the auxiliary switch **243** provided in the disconnecter **240** is manually switched to the off state to switch the disconnecter to the no-load state.

(37) Specifically, when the auxiliary switch **243** is manually turned off, power supplied to the first

main contactor **221** and the second main contactor **222** through the power line **300** is cut off, so that the first main contactor **221** and the second main contactor **222** are turned off to convert the disconnecter **240** into a no-load state.

(38) In this way, as the first main contactor **221** and the second main contactor **222** are turned off, when the disconnecter is in a no-load state, the first current cut-off switch **241** and the second current cut-off switch **242** are controlled to be turned off.

(39) Accordingly, when an abnormal operation of the ESS occurs or an inspection of the ESS is required, the disconnecter **240** may be used to block the front ends of the first main contactor **221** and the second main contactor **222**.

(40) In other words, through the above-described configurations, a function of manually blocking a current path by a conventional circuit breaker may be implemented.

(41) 1-4) First Fuse **211** and Second Fuse **212**

(42) On the other hand, the protection circuit unit of the present invention includes a first fuse in which one end is connected to the (+) end of the plurality of battery modules, and the other end is connected to the first main contactor through the disconnecter and a second fuse in which one end is connected to the (-) end of the plurality of battery modules, and the other end is connected to the second main connector through the disconnecter.

(43) The first fuse **211** and the second fuse **212** may implement a function of blocking overcurrent when an overcurrent flows in a conventional circuit breaker.

(44) 1-5) Pre-Charge FET **223**

(45) Meanwhile, the ESS **10** according to an embodiment of the present invention further includes a pre-charge FET **223** connected in parallel with the first main contactor **221** to prevent inrush current.

(46) 2. Method of Manually Blocking Current in ESS According to Embodiment of Present Invention.

(47) A method of manually blocking current in an ESS according to an embodiment of the present invention relates to a method of manually blocking current in the ESS using a disconnecter without using a circuit breaker.

(48) More specifically, by changing the structure so that the disconnecter can be turned on/off even when it is not in a no-load state, instead of an expensive circuit breaker, the present invention relates to a method of manually cutting off current in an ESS.

(49) FIG. 2 is a flowchart illustrating a method of manually blocking current in an ESS including a disconnecter according to an embodiment of the present invention.

(50) Hereinafter, a method of manually blocking current in an ESS including a disconnecter according to an embodiment of the present invention will be described with reference to FIG. 2.

(51) A method of manually blocking current in an ESS including a disconnecter according to an embodiment of the present invention includes a main contactor off step **S100** of allowing the rear end of the disconnecter to be in a no-load state, and a current path blocking step **S200** of blocking the output current path of the ESS using the disconnecter after performing the main contactor off step.

(52) Since the disconnecter can control the switch only in the no-load state, in order to manually cut off current using a disconnecter, the disconnecter should be made in a no-load state.

(53) The main contactor off step **S100** is a step of allowing the disconnecter to be in a no-load state.

(54) Specifically, the main contactor off step **S100** is a step of switching the rear end of the disconnecter into a no-load state by cutting off power supplied to the main contactor and turning off the main contactor.

(55) Meanwhile, the current path blocking step **S200** is a step of blocking the output current path of the ESS by turning off the first current cut-off switch and the second current cut-off switch of the disconnecter as the main contactor is turned off and the rear end of the disconnecter is in a no-load

state after performing the main contactor off step

(56) In this way, by sequentially performing the main contactor off step **S100** and the current path blocking step **S200**, a conventional circuit breaker can implement a function of blocking the output current path of the ESS using a disconnecter.

(57) On the other hand, although the technical idea of the present invention has been specifically described according to the above embodiment, it should be noted that the above embodiments are for the purpose of explanation and not limitation. In addition, those skilled in the art in the technical field of the present invention will be able to understand that various embodiments are possible within the scope of the spirit of the present invention.

Claims

1. An energy storage system (ESS) comprising: a plurality of battery modules; and a battery protection unit including: a microcontroller (MCU) configured to control charge and discharge of the ESS; a first main contactor and a second main contactor connecting or disconnecting between the plurality of battery modules and an output terminal of the ESS under control of the MCU; a disconnecter disposed between the first main contactor and the second main contactor and connecting or disconnecting the plurality of battery modules, the first main contactor, and the second main contactor; a first fuse having a first end connected to a positive (+) end of the plurality of battery modules and a second end connected to the first main contactor through the disconnecter; and a second fuse having a first end connected to a negative (−) end of the plurality of battery modules and a second end connected to the second main contactor through the disconnecter, wherein the first main contactor and the second main contactor are turned on or off, by the MCU, depending on whether a predetermined voltage is applied, wherein the disconnecter includes: a first current cut-off switch configured to block a current path flowing through the first main contactor; and a second current cut-off switch configured to block a current path flowing through the second main contactor, and wherein the first current cut-off switch and the second current cut-off switch are turned off when the disconnecter is in a no-load state.
2. The ESS of claim 1, wherein the first current cut-off switch has a first end connected to the positive (+) end of the plurality of battery modules and a second end connected to the first main contactor to block a current path flowing through the first main contactor, and wherein the second current cut-off switch has a first end connected to the negative (−) end of the plurality of battery modules and a second end connected to the second main contactor to block a current path flowing through the second main contactor.
3. The ESS of claim 2, wherein the disconnecter further comprises an auxiliary switch for controlling whether the predetermined voltage is applied to the first main contactor and the second main contactor.
4. The ESS of claim 3, wherein when the auxiliary switch is turned off, the first main contactor and the second main contactor are turned off by cutting off predetermined power supplied to the first main contactor and the first main contactor, and wherein when the first main contactor and the second main contactor are turned off, the ESS becomes in the no-load state.
5. The ESS of claim 4, wherein the first and second current cut-off switches are turned off after the first and second main contactors are turned off.
6. The ESS of claim 1, wherein when at least one of overvoltage, overcurrent, and high temperature is detected, the MCU is configured to turn off the first main contactor and the second main contactor.
7. The ESS of claim 1, wherein the ESS further comprises a pre-charge FET connected in parallel to the first main contactor to prevent inrush current.
8. The ESS of claim 1, wherein the first fuse is located between the first current cut-off switch and the plurality of battery modules, and wherein the second fuse is located between the second current

cut-off switch and the plurality of battery modules.

9. The ESS of claim 8, wherein the first current cut-off switch is located between the first fuse and the first main contactor, and wherein the second current cut-off switch is located between the second fuse and the second main contactor.

10. A method for performing manual power disconnection in an energy storage system (ESS) including a disconnecter, a first fuse having a first end connected to a positive (+) end of a plurality of battery modules and a second end connected to a first main contactor through the disconnecter, and a second fuse having a first end connected to a negative (−) end of the plurality of battery modules and a second end connected to a second main contactor through the disconnecter, the method comprising: a main contactor off step of converting a rear end of the disconnecter into a no-load state; and a current path blocking step of blocking an output current path of the ESS using the disconnecter after performing the main contactor off step, wherein the disconnecter includes a first current cut-off switch and a second current cut-off switch, and wherein the current path blocking step turns off the first current cut-off switch and the second current cut-off switch of the disconnecter to block the output current path of the ESS after the main contactor off step is performed and the disconnecter is in the no-load state.

11. The method of claim 10, wherein the main contactor off step turns off the main contactor by cutting off the power supplied to the main contactor, thereby converting the rear end of the disconnecter into the no-load state.
